

Name _____

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Q1. In fieldwork you often tape the circumference of a tree at breast height but need diameter (DBH) to use a height-DBH model. What is the correct conversion from measured circumference C to diameter d ?

- a. $d = C / \pi$
- b. $d = 2C / \pi$
- c. $d = C^2 / \pi$
- d. $d = \pi C$
- e. $d = \sqrt{(C/\pi)}$

Answer Q1 (A) (B) (C) (D) (E)

Q2. You are comparing two models that predict a vegetation variable Y from a sensor index X . Your priority is to minimise absolute prediction error in interpretable units of Y . Which metric should guide your choice?

- a. Adjusted R^2
- b. R^2
- c. Root Mean Squared Error (RMSE)
- d. Correlation coefficient r
- e. p-value of the slope

Answer Q2 (A) (B) (C) (D) (E)

Q3. You can easily measure DBH but not height in a forest survey. Which workflow best matches the lecture's definition of a model and the purpose of prediction?

- a. Fit a statistical model of height from DBH using a training set, then use it to predict heights for the remaining trees.
- b. Use Pythagoras with GPS coordinates to compute tree height from planimetric distances.
- c. Assume a fixed height:diameter ratio from a paper and multiply all DBHs by that constant.
- d. Measure a few heights, take the mean, and assign it to all trees.
- e. Map crowns in an equal-area projection and use crown area as height proxies.

Answer Q3 (A) (B) (C) (D) (E)

Q4. Why did the lecture emphasize using simple formulae carefully before moving into software tools?

- a. Software can only run if every formula is typed from memory.
- b. Quantitative interpretation still depends on understanding what measurements and formulae represent.
- c. Coding removes the need to think about units or measurement.
- d. Environmental science avoids mathematical calculations once data are collected.
- e. Formulae are only relevant for drawing figures, not analysis.

Answer Q4 (A) (B) (C) (D) (E)



Q5. In the lecture, what was the purpose of discussing RStudio setup before the labs?

- a. To make sure students could use the required computing environment for practical work.
- b. To replace all later lectures with self-guided programming tasks.
- c. To discourage students from using local computers.
- d. To assess students before they had seen the course material.
- e. To demonstrate that coding is separate from environmental science.

Answer Q5 (A) (B) (C) (D) (E)

Q6. Which statement best describes the role of a cloud workaround such as Google Colab in the lecture?

- a. It is the default tool for all students regardless of their computer setup.
- b. It is a backup option for students who cannot run the local software environment.
- c. It is used only for watching lecture videos.
- d. It automatically calculates all answers without coding.
- e. It is required before students can learn RStudio.

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- a. Root Mean Squared Error (RMSE)
- b. p-value of the slope
- c. Adjusted R^2
- d. Correlation coefficient r
- e. R^2

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- a. Assume a fixed height:diameter ratio from a paper and multiply all DBHs by that constant.
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- a. It is used only for watching lecture videos.
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- a. p-value of the slope
- b. Root Mean Squared Error (RMSE)
- c. R^2
- d. Adjusted R^2
- e. Correlation coefficient r

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